

AWS Cheat Sheet

Baseline: AWS CLI 2.37, CDK CLI 2.1143, SAM CLI 1.166, Kubernetes 1.36 on EKS (checked 2026-10-01). Examples use the Bookshelf scenario (\$CLUSTER=bookshelf, us-east-1, account 111122223333). Quick reference only — verify against the official docs before production use.

CHOOSING A COMPUTE SERVICE

Service	Best for	Scaling	Scale to zero	Notes
EC2 + Auto Scaling	Lift-and-shift, GPUs, OS tuning	ASG target/step/scheduled	No (ASG can be 0, you manage it)	Full control, most operations (AL2023, IMDSv2)
Lightsail	Small sites, prototypes	Manual	No	Predictable bills
ECS Express Mode	New HTTP service, fastest path	Target tracking, min/max tasks	No (min tasks stay)	Image + 2 roles; App Runner target
ECS on Fargate	Containers, sidecars, custom networking	Application Auto Scaling	No (min can be 0)	awsvpc, ALB/NLB, Service Connect
ECS on EC2 / Managed Instances	GPUs, dense packing	Capacity providers	No	No ECS control-plane fee
Lambda	Event-driven, spiky, glue code	Per-request concurrency	Yes	Outside VPC by default; INIT billed since 2025-08
EKS (Auto Mode)	Kubernetes skills/portability, platforms	HPA, KEDA, Karpenster-style	No (control-plane fee)	Pod Identity, Gateway API
App Runner	Existing services only	By concurrency	Pauses to baseline	Closed to new customers; migrate to Express Mode

MANAGED DATABASES — ENGINE → AWS OPTION

Engine	AWS option	Create (essentials)
SQL / PostgreSQL / MySQL	RDS (PostgreSQL, MySQL, MariaDB, SQL Server, Oracle, Db2); Aurora PG/MySQL; Aurora DSQL	aws rds create-db-instance --db-instance-identifier bookshelf-pg --engine postgres --engine-version 17 --db-instance-class db.m7g.large --allocated-storage 100 --storage-encrypted --master-username postgres_admin --manage-master-user-password 1 aws rds create-db-cluster --engine aurora-postgresql --serverless-v2-scaling-configuration MinCapacity=0,MxCapacity=8 + create-db-instance --db-instance-class db.serverless
NoSQL key-value	DynamoDB (on-demand default; provisioned + autoscaling)	aws dynamodb create-table --table-name Orders --attribute-definitions AttributeName=customerId,AttributeType=S ... --key-schema AttributeName=customerId,KeyType=HASH AttributeName=orderId,KeyType=RANGE --billing-mode PAY_PER_REQUEST
MongoDB	Amazon DocumentDB (instance, elastic, Serverless); Atlas on AWS; MongoDB on EKS	aws docdb create-db-cluster --db-cluster-identifier docdb-bookshelf --engine docdb --manage-master-user-password + create-db-instance
Redis	ElastiCache (Valkey, Redis OSS, Memcached; serverless or nodes); MemoryDB; Redis Cloud	aws elasticache create-serverless-cache --serverless-cache-name cache-bookshelf --engine valkey
Elasticsearch / Solr	OpenSearch Service (domains or Serverless; fork, not Elasticsearch); Elastic Cloud; ECK on EKS; Solr self-managed	aws opensearch create-domain --domain-name bookshelf-search aws opensearchserverless create-collection --name books --type SEARCH
Partner / graph / metrics	Couchbase Capella/Operator; Neo4j AuraDB/Helm (Neptune is a different engine); Qdrant Cloud/Helm; Managed Prometheus	aws amp create-workspace --alias bookshelf aws neptune create-db-cluster --engine neptune

Auth: IAM database authentication (aws rds generate-db-auth-token); managed master passwords live in Secrets Manager; RDS Proxy pools Lambda connections; Aurora Serverless v2 scales to 0 ACU; DynamoDB PITR via update-continuous-backups.

HIERARCHY & CREDENTIALS

Organization → root → OUs → accounts → resources. SCPs/RCPs limit, never grant. Region = independent geography; AZ = isolated data centers within it (use ≥2). Root user: MFA, then put it away. Daily access: IAM Identity Center permission sets give short-lived role credentials. Prefer roles over users.

CLI ESSENTIALS

```
aws configure sso
aws sso login --profile bookshelf-dev
aws login # console credentials, CLI 2.32+
aws sts get-caller-identity
aws configure list-profiles
aws configure set region eu-west-1
aws ec2 describe-regions --output text \
  --query "Regions[0].RegionName"

--output json|table|text|yaml; --query is JMESPath.
laC: aws cloudformation deploy, sam deploy, cdk deploy.
```

ECS & FARGATE

```
aws ecs create-express-gateway-service \
  --service-name catalog-api \
  --execution-role-arn <exec-role> \
  --infrastructure-role-arn <infra-role> \
  --primary-container image=curi,containerPort=8080
aws ecs register-task-definition \
  --cli-input-json file://td.json
aws ecs create-service --cli-input-json file://sv.json
aws ecs update-service --cluster bookshelf \
  --service catalog-api --force-new-deployment
aws logs tail /ecs/bookshelf/catalog-api --follow
```

Roles: execution role (image, logs, secrets), task role (app permissions), infrastructure role (ALB/volumes). Built-in blue/green (2025-07), linear/canary (2025-10). Application Auto Scaling target tracking; Fargate Spot.

LAMBDA

```
aws lambda create-function --function-name f \
  --runtime python3.14 --handler app.handler \
  --role <role-arn> --zip-file fileb://f.zip
--name live --function-version 5
aws lambda put-provisioned-concurrency-config \
  --function-name f --qualifier live \
  --provisioned-concurrent-executions 5
```

Execution role (what it may do) vs. resource policy (who may invoke). Async payload 1 MB. Function URLs, layers, SnapStart, event source mappings (SQS, Kinesis, DynamoDB streams).

ECR & EKS

```
aws ecr get-login-password | docker login \
  --username AWS --password-stdin $REGISTRY
docker push $REGISTRY/bookshelf/catalog-api:1.4.2
eksctl create cluster --name $CLUSTER --enable-aws-iam-auth --kubernetes-version 1.36
aws eks update-kubeconfig --name $CLUSTER
aws eks create-access-entry --cluster-name $CLUSTER \
  --principal-arn <arn>
aws eks create-pod-identity-association \
  --namespace bookshelf --service-account-notifier \
  --role-arn <arn>
aws eks update-cluster-version --name $CLUSTER \
  --kubernetes-version 1.36
```

Auto Mode vs. managed node groups vs. Fargate. Access entries replace aws-auth; Pod Identity vs. IRSA. Load Balancer Controller serves Ingress or Gateway API. Run aws eks list-insights before upgrades; old versions add extended-support fees.

S3

```
aws s3api create-bucket --bucket b --region us-east-1
aws s3 cp f s3://b/k --sse aws:kms
aws s3 presign s3://b/k --expires-in 600
aws s3api put-bucket-lifecycle-configuration ...

Classes: Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant/Flexible/Deep Archive, Express One Zone. Strong consistency; SSE-S3 default; ACLs off; Block Public Access on; SSE-C blocked (2026-04). Sites: private bucket + CloudFront OAC. Empty versioned buckets before deleting.
```

SECRETS, PARAMETERS & KMS

```
aws secretsmanager create-secret --name app/db \
  --kms-key-id alias/bookshelf-data --secret-string '{"...}'
aws secretsmanager get-secret-value --secret-id app/db
aws ssm put-parameter --name /app/prod/page-size \
  --type String --value 25
aws kms enable-key-rotation --key-id $KEY

ECS secrets[0].valueFrom takes a Secrets Manager or Parameter Store ARN; on EKS use ASCP with Pod Identity. Secrets Manager rotates and replicates; Parameter Store suits plain config. KMS deletion waits 7–30 days.
```

CLOUDFRONT

```
aws cloudfront create-origin-access-control \
  --origin-access-control-config Name=oac,SigningProtocols=igv4,SigningBehavior=always,OriginAccessControlOriginType=s3
aws cloudfront create-distribution \
  --distribution-config file://d.json
aws cloudfront create-invalidation \
  --distribution-id $ID --paths /*/*

Bucket policy allows CloudFront.amazonaws.com with AWS:SourceArn. Cache, origin request and response-headers policies. ACM certificate in us-east-1. Flat-rate plans (2025-11), VPC origins, WAF, key groups.
```

APPCONFIG / CONFIGMAPS

```
aws appconfig start-deployment \
  --application-id $A --environment-id $E \
  --configuration-profile-id $P --configuration-version 1 \
  --deployment-strategy-id AppConfig.Canary10Percent20Minutes
aws appconfigdata get-latest-configuration ...
kubectrl create configmap app-config --from-file=app.yaml
kubectrl rollout restart deployment/catalog-api

Freeform or feature-flag profiles with validators and rollback alarms. AppConfig Agent caches locally (Lambda layer, ECS sidecar); Spring Cloud AWS 4.x reads AppConfig. ConfigMap changes need a rollout restart.
```

VPC

```
aws ec2 create-vpc --cidr-block 10.0.0.0/16
aws ec2 create-subnet --vpc-id $V \
  --cidr-block 10.0.1.0/24 --availability-zone us-east-1a
aws ec2 authorize-security-group-ingress \
  --group-id $TASK_SG --protocol tcp --port 8080 \
  --source-group $ALB_SG
aws ec2 create-vpc-endpoint --vpc-endpoint-type Gateway \
  --service-name com.amazonaws.us-east-1.s3 ...

Three tiers: public (ALB), private app, private data. Security groups are stateful and can reference each other; NACLs are stateless. Public IPv4 and NAT gateways cost money; S3/DynamoDB gateway endpoints are free.
```

LOAD BALANCERS & API GATEWAY

Decision: HTTP/S rules, JWT, WAF → ALB; TCP/UDP/TLS, static IPs → NLB; inline appliances → GWLB; global anycast → Global Accelerator; cross-VPC services → VPC Lattice.

```
aws elbv2 create-listener --load-balancer-arn $ALB \
  --protocol HTTPS --port 443 --certificates CertificatesArn=$C \
  --default-actions Type=forward,TargetGroupArn=$TG
aws apigatewayv2 create-api --name bookshelf \
  --protocol-type HTTP

API Gateway: HTTP API (cheap, JWT, VPC link), REST API (keys, usage plans, WAF, caching), WebSocket. Route key "GET /books/{id}"; custom domain with ACM + Route 53.
```

EMAIL (SES)

```
aws sesv2 create-email-identity --email-identity-example.com
aws sesv2 put-account-details --mail-type TRANSACTIONAL \
  --website-url https://www.example.com \
  --use-case-description "..." --contact-language EN
aws sesv2 send-email \
  --from-email-address no-reply@example.com \
  --destination ToAddresses=a@example.org \
  --content '{"Simple":{...}}'

Publish DKIM CNAMEs, SPF and DMARC in Route 53. Sandbox: verified recipients only; leave it with put-account-details. Handle bounces and complaints via event destinations and the suppression list. Use the task or Lambda role (ses:SendEmail), SMTP credentials only for legacy apps.
```

MONITORING

```
aws cloudwatch put-metric-alarm --alarm-name api-5xx \
  --namespace AWS/ApplicationELB \
  --metric-name HTTPCode_Target_5XX_Count \
  --statistic Sum --period 60 --evaluation-period 3 \
  --threshold 5 --comparison-operator GreaterThanThreshold \
  --alarm-actions $SNS_ARN
aws logs tail /aws/lambda/f --follow --since 10m

Logs Insights: filter @message like /ERROR/ | stats count() by bin(5m). Traces: OpenTelemetry/ADOT with CloudWatch Application Signals. CloudTrail trail to S3 for audit; Config, Security Hub, GuardDuty for governance.
```

COST & SUPPORT

```
aws budgets create-budget --account-id $A \
  --budget file://budget.json
aws ce get-cost-and-usage --granularity MONTHLY \
  --time-period Start=...,End=... \
  --metrics UnblendedCost \
  --group-by Type=DIMENSION,Key=SERVICE

Budgets, anomaly detection, cost-allocation tags. Savings Plans, Reserved, Spot. Free plan (accounts since 2025-07-15): credits for 6 months, then Paid plan. Support: Basic, Business Support+, Enterprise, Unified Operations. Delete what you create: NAT gateways, ALBs, public IPv4, EBS.
```

VPN

```
aws ec2 create-client-vpn-endpoint \
  --client-cidr-block 172.16.0.0/22 \
  --server-certificate-arn $CERT \
  --authentication-options Type=certificate-authentication,MutualAuthentication=(ClientRootCertificateChainArns=$CA) \
  --connection-log-options Enabled=false
aws ec2 create-vpn-connection --type ipsec.1 \
  --customer-gateway-id $CGW --vpn-gateway-id $VGW

Client VPN: mutual TLS, Active Directory or SAML. Site-to-Site: two tunnels per connection, virtual private or transit gateway, BGP. Verified Access for per-app zero trust; SSM Session Manager replaces bastions.
```

RETIREMENTS & RENAMES — CHECK DATES BEFORE RELYING ON THEM

- Amazon Linux 2: end of support 2026-06-30; use AL2023.
- Aurora Serverless v1: end of life 2025-03-31; use Serverless v2.
- App Runner: closed to new customers 2026-04-30; use ECS Express Mode.
- Cognito CLI: end of support 2026-06-12.
- CloudFront closed to new customers since 2024-07. CodeCatalyst: closed 2025-11-07. CodeCommit: back in general availability 2025-11-24.
- X-Ray SDKs/daemon: maintenance mode from 2026-02-25 (no hard end-of-support date on the official page; earlier AWS communication gave 2027-02-25); use OpenTelemetry.
- Pinpoint: end of support 2026-10-30. WorkMail: 2027-03-31. App Mesh: 2026-09-30 (use VPC Lattice).
- Support plans: Developer, Business and Enterprise On-Ramp discontinued 2027-01-01.
- CLI v1: maintenance 2026-07-15, end of support 2027-07-15. JS SDK v2: ended 2025-09-08. Java SDK v1: ended 2025-12-31.
- Lambda: python3.10 deprecated 2026-10-31; python3.8/3.9 already deprecated.
- S3 Object Lambda, Snowball Edge, Migration Hub: closed to new customers 2025-11-07. Glacier vaults: closed 2025-12-15.
- Postgres 13 left standard support 2026-02-28; MySQL 8.0 2026-07-31 (Extended Support fees follow). Redis OSS 6: 2027-01-31.
- Timestream LiveAnalytics: closed 2025-06-20. QLDB: ended 2025-07-31.
- Kendra: closed to new customers 2026-07-30. CloudTrail Lake: closed 2026-05-31.
- Renames: Elasticsearch Service → OpenSearch Service; AWS SSO → IAM Identity Center; Kinesis Data Firehose → Amazon Data Firehose; QuickSight → Amazon Quick; Application Migration Service → AWS Transform MGN; OAI → Origin Access Control.